J.A.R.V.I.S. — OS HARDWARE & INFRASTRUCTURE FUNDING PROPOSAL

EDGE AI DEPLOYMENT FOR RAY-BAN SMART GLASSES & ON-DEVICE LOCAL MODELS

SUBMISSION DEADLINE: NOVEMBER 11, 2026 | PROJECT VERSION: 4.2

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1. EXECUTIVE SUMMARY

JARVIS-OS is a custom, voice-first Linux operating system tailored for head-mounted smart glasses and autonomous edge devices. Unlike cloud-tethered voice assistants, JARVIS-OS operates offline-first using on-device neural processing, real-time optical gesture tracking, and full-duplex WebRTC communications. This funding proposal requests **$1,896 USD** to acquire dedicated edge hardware—specifically the **Google Pixel 10a** for local on-device neural model inference, Ray-Ban smart glasses optics, and standalone cellular telemetry.

2. ITEMIZED HARDWARE & INFRASTRUCTURE BUDGET

Item #	Hardware Component / Service	Technical Purpose in JARVIS-OS	Est. Cost (USD)
1	**Google Pixel 10a (16GB RAM)**	Dedicated host device with Tensor NPU. Runs local quantized LLMs (Llama 3.2, Dolphin, Gemini Nano) for 100% offline edge inference with zero cloud token bills.	$499.00
2	**Ray-Ban Smart Glasses Hardware**	Head-mounted optical sensors, dual spatial microphones, and high-contrast transparent HUD optics for wearable hands-free glass interface.	$379.00
3	**Quectel 5G/LTE Cellular Modem**	Standalone cellular telemetry module enabling real direct phone calls to mobile numbers (e.g. 8825586080) and independent remote data connectivity.	$189.00
4	**NVIDIA Jetson Orin Nano (8GB)**	Edge development board for continuous computer vision, 30 FPS hand gesture tracking, and real-time point-cloud spatial decomposition.	$499.00
5	**WebRTC & STUN/TURN Audio Server**	Low-latency audio relay server for real full-duplex VoIP/cellular audio streaming between glasses, tablet, and host PC.	$150.00
6	**High-Capacity 24,000mAh Battery & Rig**	PD 65W portable power bank, low-profile high-flex USB-C cabling, and passive thermal heatsinks for wearable field deployment.	$180.00
	TOTAL REQUESTED FUNDING	**Complete edge hardware package & telecommunications stack**	**$1,896.00**

3. TECHNICAL JUSTIFICATION: GOOGLE PIXEL 10a FOR LOCAL AI

The **Google Pixel 10a** is uniquely suited as the pocket compute brick for JARVIS-OS smart glasses for three core reasons:
• **Hardware NPU Acceleration:** The Google Tensor processing unit provides native INT4/INT8 matrix multiplication, delivering sub-second token generation for local LLMs without draining battery.

• **Zero Cloud Latency & Cost:** Running models locally on the Pixel 10a eliminates API usage costs and cloud credit burn completely, guaranteeing privacy and 100% offline capability.
• **Form Factor & Thermal Efficiency:** The lightweight profile allows it to operate in a pocket while tethered via ultra-low latency Wi-Fi 7/Bluetooth 5.4 to the Ray-Ban smart glasses.

4. ROADMAP & MILESTONES (DEADLINE: NOVEMBER 11, 2026)

Date	Phase & Deliverables	Target Metric / Output
Nov 4 - 6	**Local Model Tuning on Edge Device** Deploy quantized Dolphin-Llama3 and Gemini Nano directly on host hardware.	100% offline response <1.0s, 0 cloud API dependencies.
Nov 7 - 8	**Full-Duplex WebRTC Voice Pipeline** Integrate real WebRTC audio bridge with telephone dialing to +91 8825586080.	Two-way voice stream with audible ringing and <180ms latency.
Nov 9 - 10	**Optical Gesture & Glass HUD Field Test** End-to-end testing with 12 optical hand gestures and transparent glass HUD.	Robust 30 FPS gesture recognition under natural lighting.
Nov 11	**FINAL RELEASE & INVESTOR DEMONSTRATION** Live presentation of complete autonomous JARVIS-OS running on smart glasses.	**Funding Milestone Met: Production Ready.**

Prepared by: JARVIS-OS Core Development Team
Project Repository: E:\jarvis | Autonomous System Architecture v4.2
Target Hardware: Ray-Ban Smart Glasses + Google Pixel 10a Edge Engine